TEMASEK JUNIOR COLLEGE
Preliminary Examinations
Higher 1

BIOLOGY 8875/01

Paper 1 Multiple Choice **21st September 2011**

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

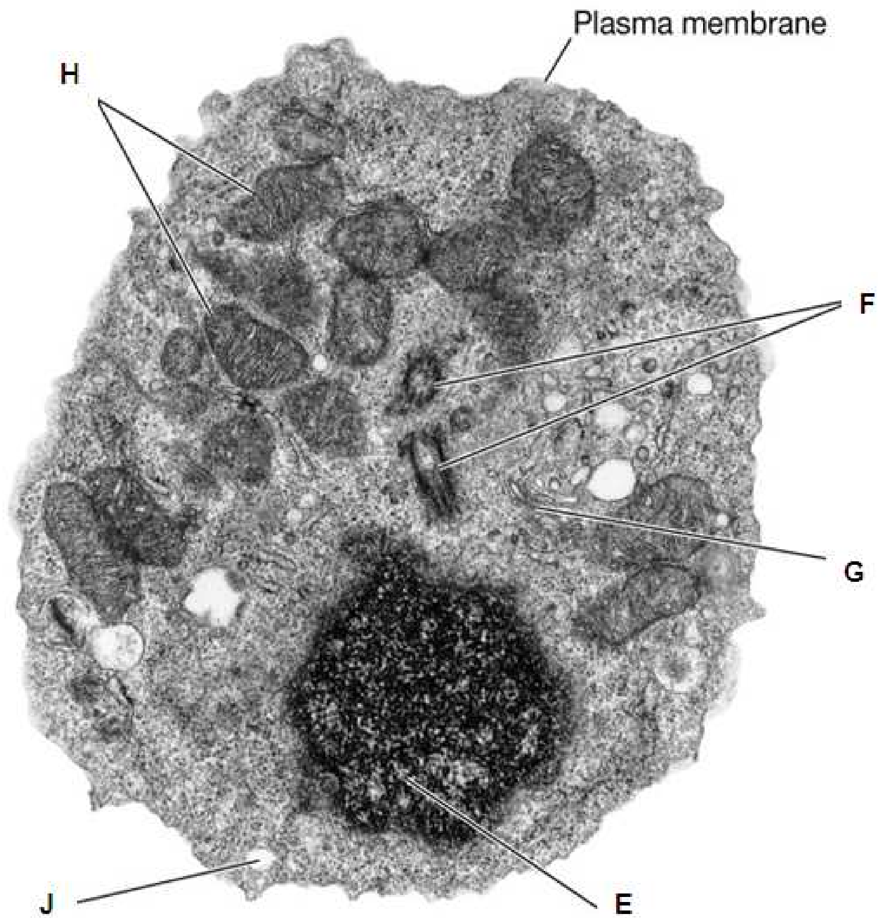
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

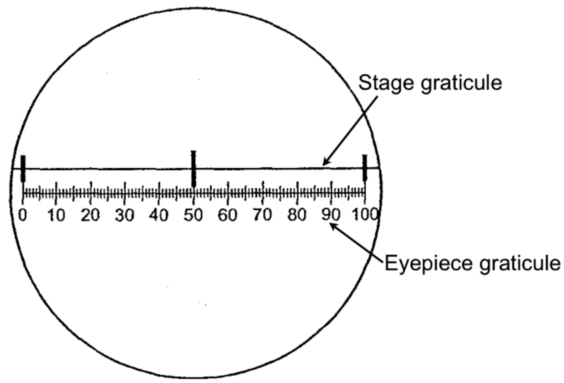
1 An electron micrograph of a cell is shown below.

Match the organelles E, F, G, H and J associated with the cellular processes listed.



	E	F	G	H	J
A	DNA replication	Digestion of material	Organizes the spindle	Oxidative phosphorylation	Packaging of secretory products
B	Oxidative phosphorylation	Organizes the spindle	Digestion of material	DNA replication	Packaging of secretory products
C	Organizes the spindle	Digestion of material	Oxidative phosphorylation	Packaging of secretory products	DNA replication
D	DNA replication	Organizes the spindle	Packaging of secretory products	Oxidative phosphorylation	Digestion of material

2 The figure below shows a stage graticule viewed through a light microscope fitted with an eyepiece graticule. The stage graticule has divisions of 0.1 mm.



The same microscope is then used to estimate the diameter of a plant cell, which measures approximately 67 smallest divisions on the eyepiece graticule.

What is the approximate diameter of the plant cell?

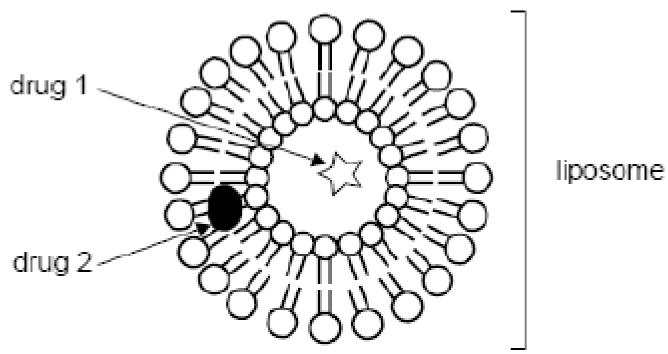
A 0.13 µm

B 1.3 µm

C 13 µm

D 130 µm

3 Liposomes can be found in the cytosol and are formed from a double layer of phospholipid molecules identical to those found in plasma membranes. Liposomes can also be manufactured and used to carry medicinal drugs into cells.



By examining the figure above, it is reasonable to expect that

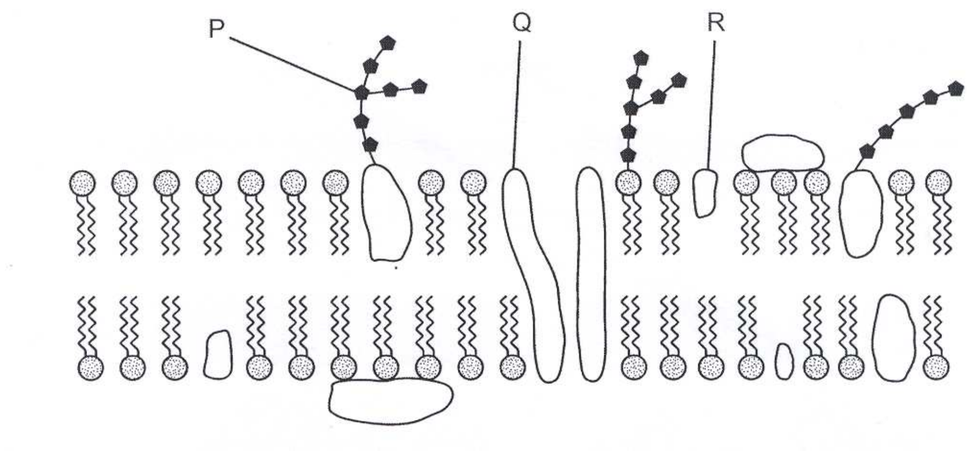
A drug **1** is lipophilic.

B drug **2** is water soluble.

C the interior of the liposome is aqueous.

D liposome-based membranes have a rigid structure.

4 The diagram below shows part of a cell surface membrane.



What is the correct function for each of the structures labelled?

	Regulates membrane fluidity	Forms hydrogen bonds with water to stabilise membrane	Transports ions and large polar molecules
A	P	R	Q
B	P	Q	R
C	Q	R	P
D	R	P	Q

5 Which of the following reactions illustrates hydrolysis?

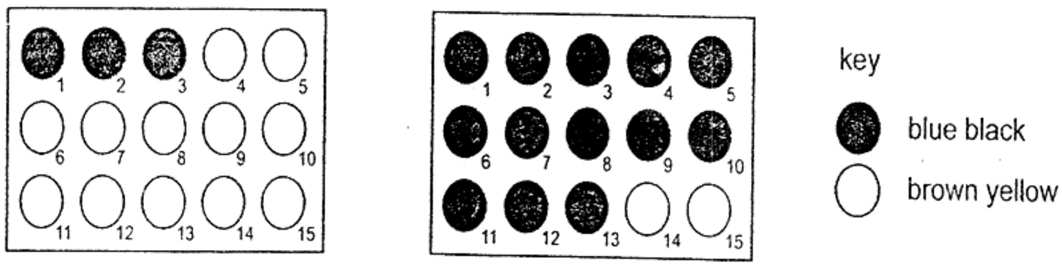
A The reaction of two monosaccharides to form a disaccharide with the release of water.

B The breakdown of a nucleotide to form a phosphate, a ribose sugar and a nitrogenous base with the production of a molecule of water.

C The synthesis of two amino acids to form a dipeptide with production of one molecule of water.

D The breakdown of a triglyceride molecule to give glycerol and fatty acids with the utilization of three molecules of water.

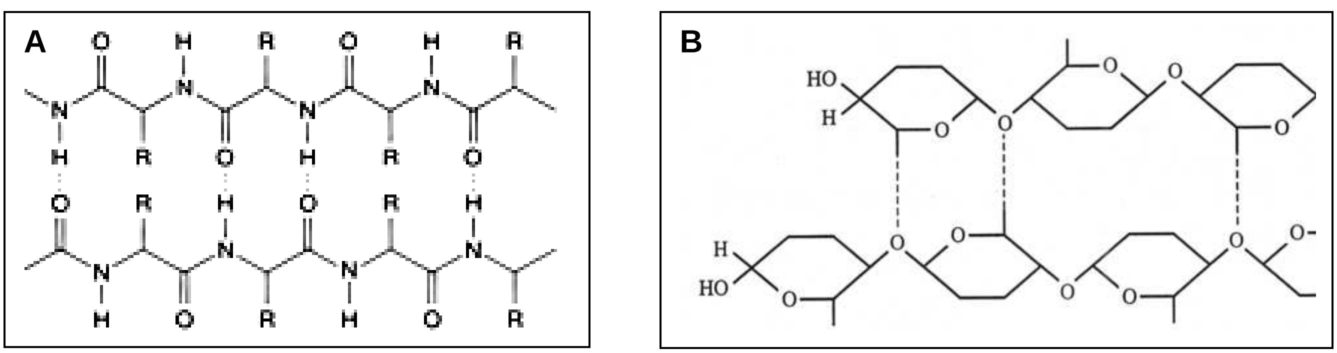
6 An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.



Which of the following shows the correct temperatures and times for the complete digestion of starch?

	temperature/ ^oC	time/s
A	30	3.15
	10	0.45
B	30	45
	10	195
C	10	45
	30	195
D	10	3.15
	30	0.45

7 The following shows two molecules that perform structural functions.



Which of the following statements does not describe a similarity between molecules A and B?

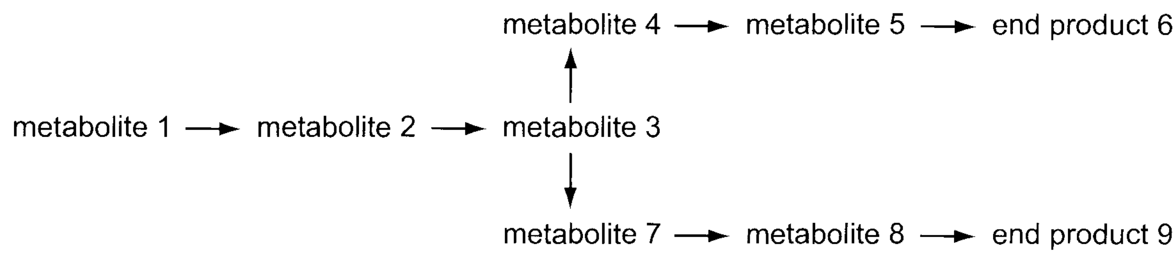
A Hydrogen bonds form cross-linkages between neighbouring chains.

B Monomers are joined together via peptide bonds.

C Monomers can be extracted by hydrolysing the bonds within each chain.

D Covalent bonds are present in both molecules.

8 The diagram shows a metabolic pathway controlled by end product inhibition. As the concentration of the end product increases above a set level, an enzyme in the pathway leading to the end product is inhibited.



An increase in the concentration of end product 6 does not lead to a decrease in the synthesis of end product 9.

Which enzyme is inhibited by end product 6?

A metabolite 1 → metabolite 2

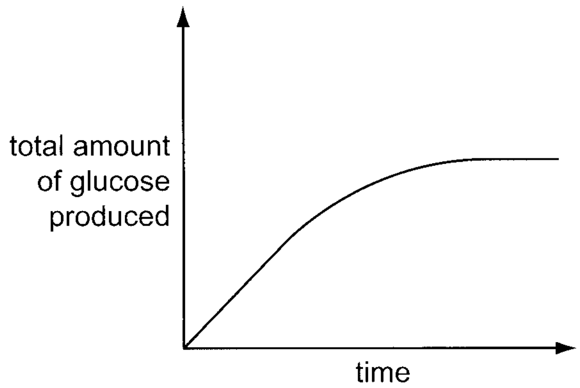
B metabolite 2 → metabolite 3

C metabolite 3 → metabolite 4

D metabolite 3 → metabolite 7

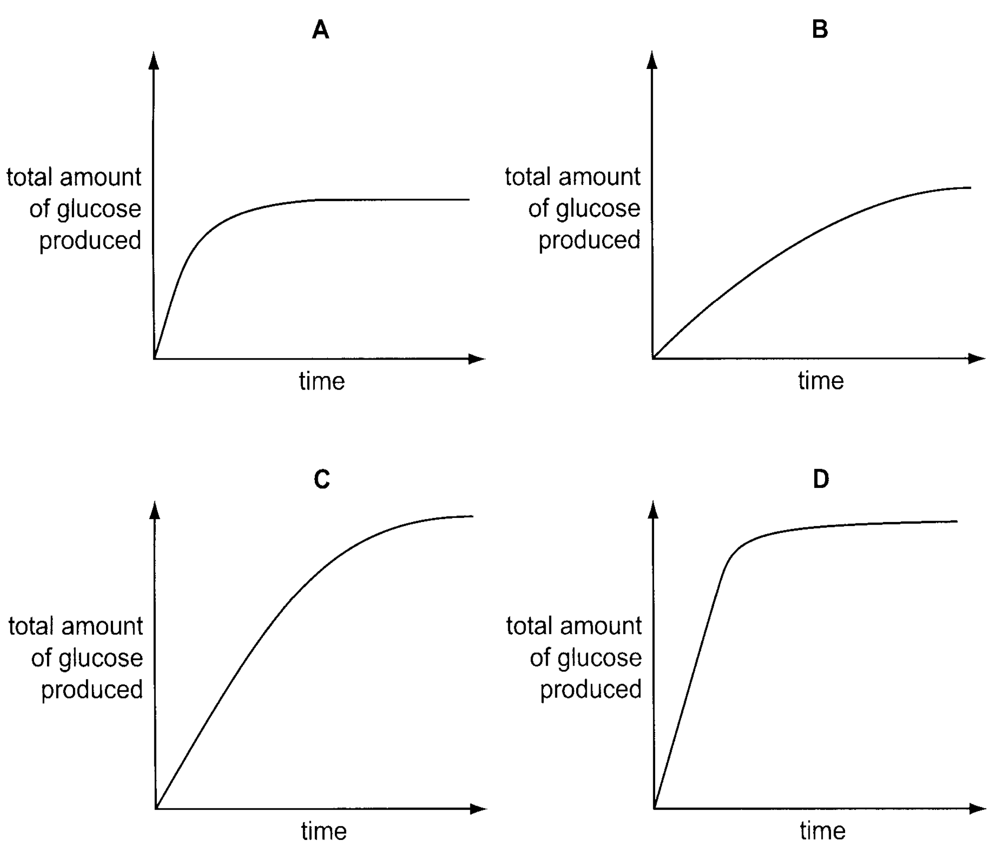
9 Lactose is a disaccharide present in milk. The enzyme β galactosidase catalyses the conversion of lactose to glucose and galactose.

10 cm^3 of a 1% β galactosidase solution was added to 10 cm^3 of milk. The graph shows the total amount of glucose produced over the next ten minutes.



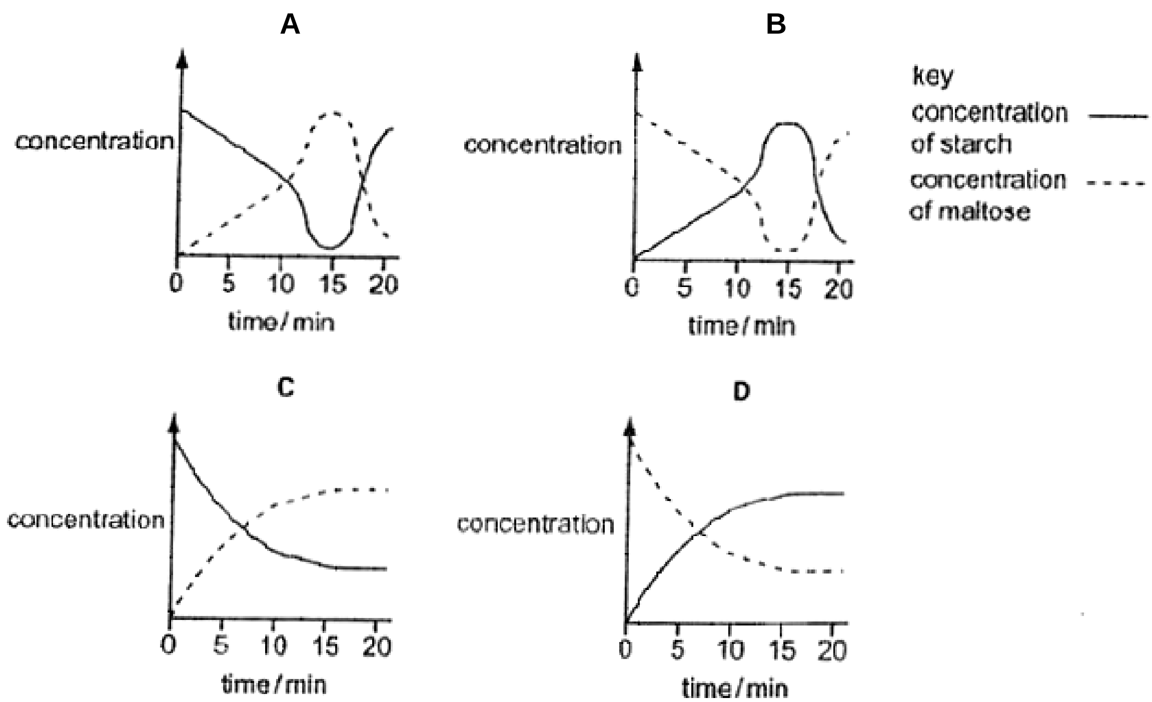
In another experiment, 10 cm^3 of a 2% β galactosidase solution was added to 10 cm^3 of milk.

Which graph shows the results that would be obtained?

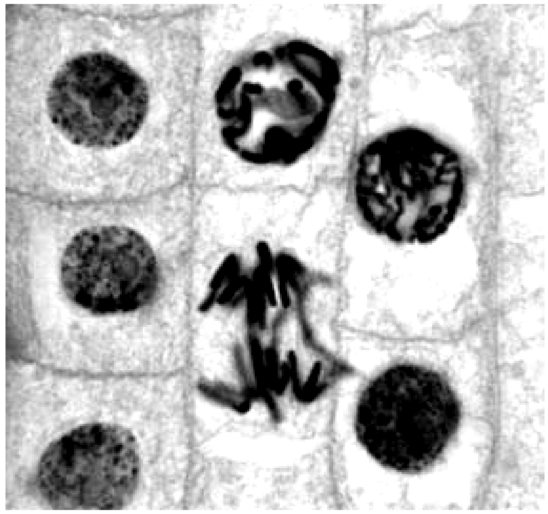


10 In an experiment using starch and amylase, the concentrations of starch and maltose present in the reacting mixture are measured every minute for 20 minutes. 1% hydrochloric acid is added after 10 minutes and the mixture is heated to 60°C at 14 minutes.

Which graph represents the results of this experiment?



11 The figure below shows a micrograph of a few onion root tip cells undergoing cell division.



Which of the following statement(s) is/are true?

I The resulting daughter cells will contain half the number of chromosomes.

II None of the cells are currently undergoing prophase.

III Sister chromatids have separated in at least one of the cells.

IV Spindle microtubules have invaded the nuclear region in most of the cells.

A III only

B III and IV

C I, II and IV

D I, III and IV

12 If, in mitotic division, homologous chromosomes finish the division in the same nucleus, this is

A an example of non-disjunction

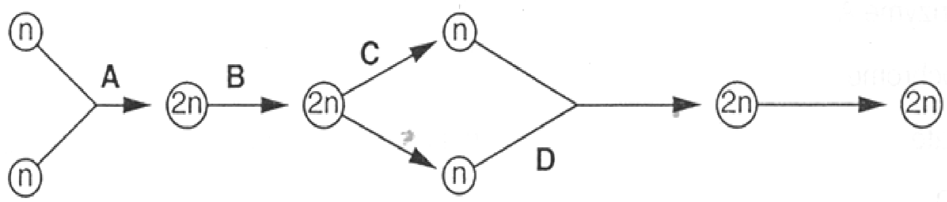
B completely normal

C a form of polyploidy

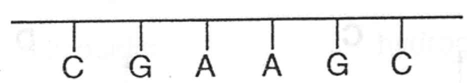
D 50% probable

13 The diagram represents the life cycle of an animal.

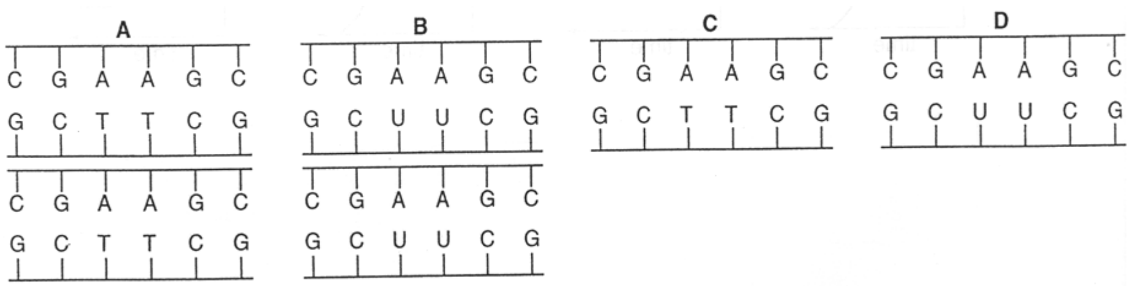
At which stage in the life cycle does mitosis occur?



14 The diagram represents one strand of part of a DNA molecule.



Which of the following would this section and its complementary strand give rise to after semi-conservative replication?



15 A gene has the following sequence on the template strand:

1	5	10	15

3' – C T G G C A T G C C T T C G G A A A -5'

Which of the following would probably result in the least change in the activity of the protein coded by this gene?

A Substitution of A for G in position 3

B Deletion of C at position 5

C Deletion of A at position 16

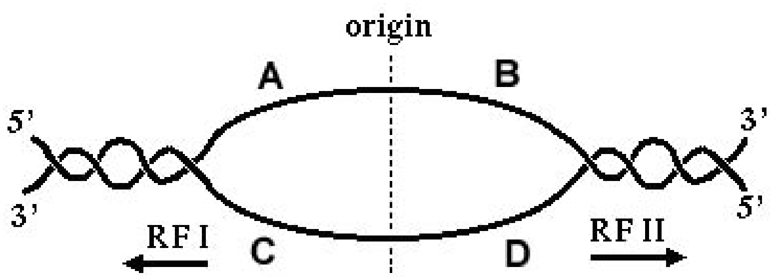
D Addition of G between positions 14 and 15

[Turn over

16 A new life form discovered on Mars has a genetic code similar to those on Earth, except there are five different DNA bases and the sequences are translated as doublets instead of triplets. How many different codons can be specified by this genetic code?

A	5	**B**	25	**C**	32	**D**	64

17 In the figure below, which DNA segment would serve as a template for continuous synthesis with DNA polymerase proceeding in the direction of replication fork I (RFI)?



18 In a laboratory, the 5' 7-methylguanosine cap and the length of the poly A tail of an mRNA transcript are altered in order to examine their effects on the number of functional proteins translated.

mRNA	**Alteration**
1	5' cap removed, 1000-adenine tail intact
2	5' cap removed, 500-adenine tail intact
3	5'cap present with 500-adenine tail
4	5'cap present with 300-adenine tail

Using the information given in the table above, predict the relative amount of protein produced in the descending order.

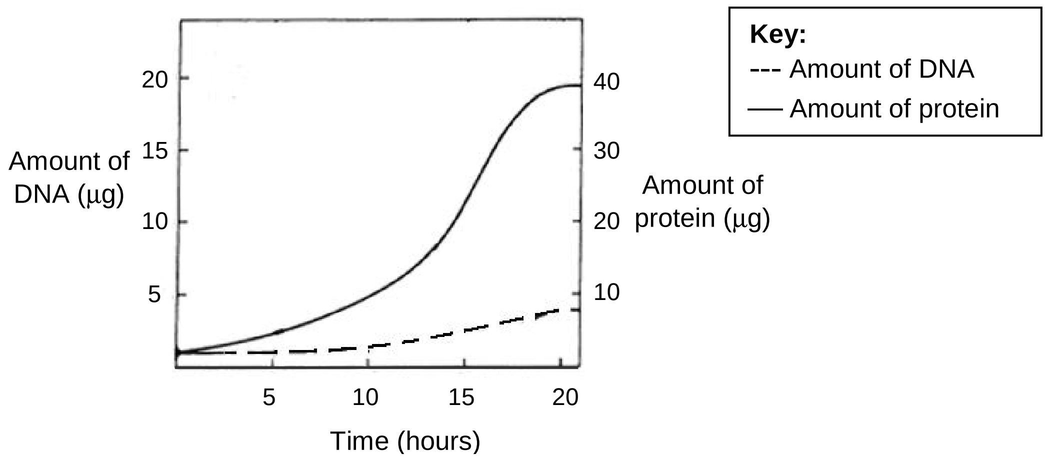
A 1,3,2,4

B 1,2,3,4

C 3,1,4,2

D 3,4,1,2

19 During egg production in *Drosophila*, there is an increased production of eggshell proteins. The genes which code for the eggshell proteins are clustered on the X chromosome.



Which of the following best explains the trends observed in the graph?

A Gene amplification has occurred, increasing the rate of transcription. Cytoplasmic poly(A) tail addition has occurred, increasing the duration of translation.

B DNA replication has occurred via the rolling circle mechanism, increasing the rate of and duration of translation.

C Gene amplification has occurred, increasing the rate of mutation. Cytoplasmic poly(A) tail addition has occurred, increasing the duration of translation.

D DNA replication has occurred. The abundance and activity of eukaryotic translation initiation factors are increased simultaneously, thereby increasing the rate of translation.

20 Cancer cells divide out of control, forming tumours.

Which statement describes the difference between a cancer cell and a normal cell?

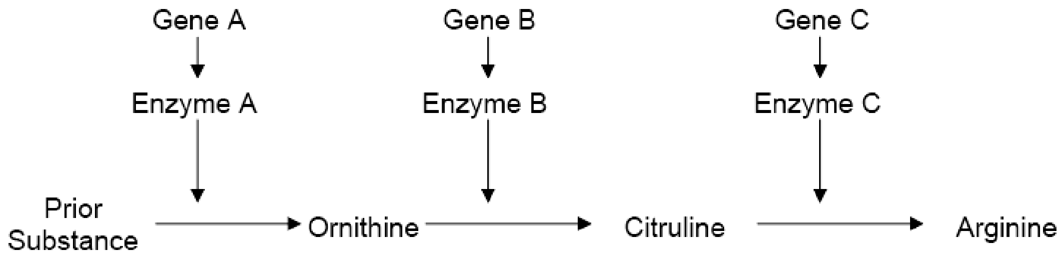
A Cancer cells do not undergo cytokinesis

B Cancer cells have a shorter interphase

C Cancer cells do not have metaphase

D Only cancer cells have mutated DNA

21 The bread mould, *Neurospora*, normally produces its own amino acids (ornithine, citruline and arginine) from raw materials through a system of enzymes.



If a nutritional mutant of *Neurospora* can only survive when provided with citruline, the mutation probably affected the following gene which could

A only be gene **A**

B only be gene **B**

C only be gene **C**

D be either gene **A** or **B**

22 Which statement about anaerobic respiration is correct?

A Animals are unable to use lactate for the production of ATP.

B From one molecule of glucose, ethanol and lactate production yield the same amounts of ATP.

C From one molecule of glucose, ethanol production yields more energy than lactate production.

D Yeast is able to respire ethanol for the production of ATP.

23 In light, Mg^{2+} ions move into the stroma. In the dark, the concentration of Mg^{2+} ions in the stroma is low.

In high Mg^{2+} concentrations, CO_2 and Mg^{2+} react with the active site of RuBP carboxylase (rubisco), making a carbamate group ($C0_2NH$) and making the rubisco active. In low Mg^{2+} concentrations the carbamate group dissociates making the rubisco inactive.

Inactive rubisco will bind tightly at its active site with any RuBP present, so that no rubisco catalysis can take place. An ATPase enzyme called rubisco activase can, in high concentrations of Mg^{2+}, release the RuBP from the rubisco, producing a carbamate group and activating the rubisco.

Which statement is supported by these facts?

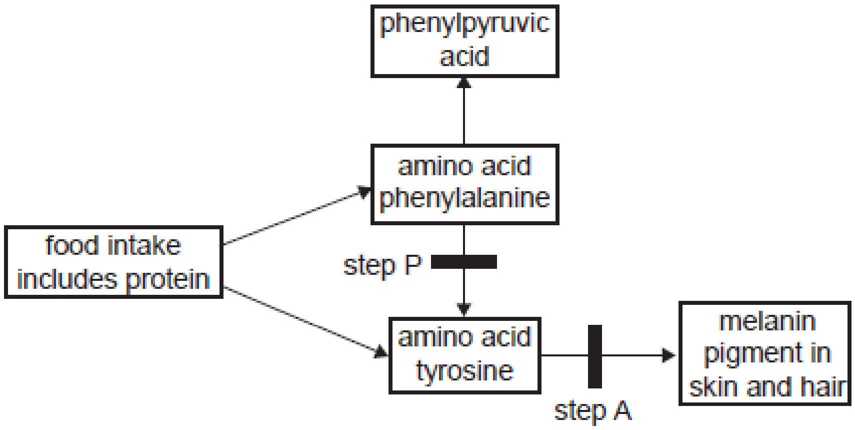
A Inactivated rubisco can be reactivated by NADPH made in the light-depending reaction.

B Low concentrations of 3C compound in the stroma of the chloroplast deactivate rubisco.

C Rubisco without the carbamate group is inhibited by low concentrations of RuBP.

D The rate of carbon dioxide fixation increases when the carbamate group dissociates from rubisco.

24 The figure below shows part of a metabolic pathway involving the amino acids phenylalanine and tyrosine. One or more enzymes are involved at each step in such a pathway.



Two steps have been highlighted, step **P** and step **A**. Each is under the control of a single gene.

Step **P** is controlled by **gene P** which has the alleles

P : production of phenylalanine hydroxylase

P’ : no enzyme

Step **A** is controlled by **gene A** which has the alleles

A : production of tyrosinase

a : no enzyme

A couple, each heterozygous at the **P** and **A** loci, have a daughter Emily.

With reference to these two genes, what is the chance that Emily has the same genotype as her parents?

A 0% **B** 25% **C** 75% **D** 100%

25 The fur colour of rabbits is controlled by a gene with 3 alleles. The phenotypes are black, brown and white fur. 4 crosses were repeated many times. The crosses and the outcomes of these crosses are shown in the table below.

Cross	**Parents**	**Offspring phenotype and ratio**
1	black x black	3 black : 1 white
2	brown x white	1 brown : 1 white
3	black x black	3 black : 1 brown
4	white x white	all white

From the data, it is possible to conclude that

A brown fur is recessive to white fur.

B all of the white fur offspring are heterozygous.

C two thirds of the black fur offspring in cross **3** are heterozygous.

D the black fur parents in cross **1** have the same genotype as the black fur parents in cross **3**.

26 Approximately 1 in 20 Europeans are heterozygous for a recessive allele responsible for the genetic condition (CF). People who are homozygous for CF have reduced life expectancy. Heterozygotes are more resistant to some bacterial infections of the gut, such as typhoid fever, than homozygotes for the normal, dominant allele.

What could explain the high incidence of the recessive CF allele in the European population?

A natural selection favouring heterozygotes

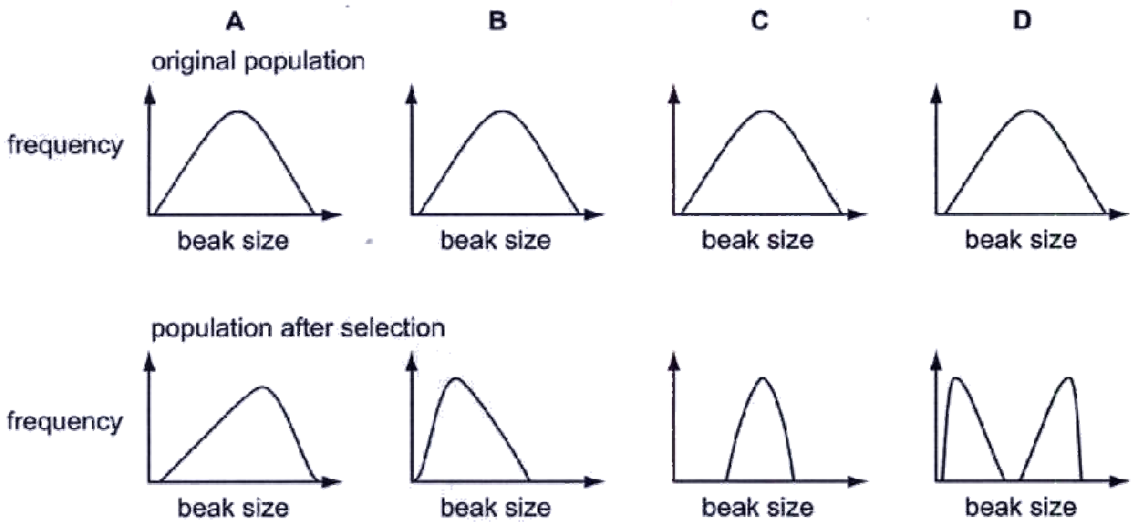
B natural selection favouring homozygotes for the recessive CF allele

C lack of genetic drift in the European population

D mutation

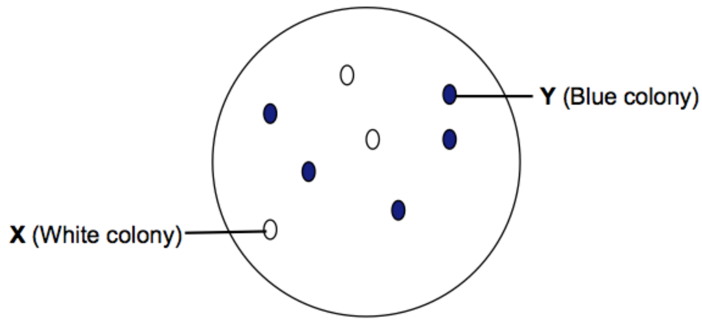
27 One species of finch on an island shows variation in beak size. Birds with larger beaks can eat larger seeds. Following a dry period, larger seeds and nuts are more plentiful than small seeds.

Which graph shows the effect of natural selection on beak size of these birds following a dry period?



28 pGEM-T is a 7.25kb plasmid cloning vector often used in creating recombinant DNA. It contains two selectable marker genes, *TetR* and *lac*Z. The multiple cloning site (MCS) of the cloning vector is located within the *lac*Z gene.

Using *Sac*II as the restriction enzyme, a 2.31kb insert (gene of interest) is cloned into the MCS of plasmid pGEM-T and the recombinant plasmid is transformed into *E. coli* cells. The transformants are then plated on medium containing tetracycline and X-Gal. The following result is obtained.



Colonies **X** and **Y** are isolated and the plasmid DNA from the colonies is extracted. The extracted plasmids are then digested with *Sac*II. Which of the following correctly shows the fragments obtained?

	Colony X	Colony Y
A	A 9.56kb fragment	A 7.25kb fragment
B	A 7.25kb fragment	A 9.56kb fragment
C	7.21kb and 0.34kb fragments	7.21kb and 2.31kb fragments
D	7.21kb and 0.34kb fragments	7.21kb and 0.34kb fragments

29 A multiple cloning site

A contains many copies of a cloned gene.

B allows flexibility in the choice of restriction enzymes for cloning.

C allows flexibility in the choice of organism for cloning.

D contains many copies of the same restriction enzyme site.

30 Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon.

Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as genetically modified?

A Salmon providing eggs for treatment

B Trout providing reproductive organ tissue

C Young triploid salmon

D Young trout